

Derbyshire Shared Care Pathology Guidelines

Hypothyroidism

Purpose of Guideline

The investigation, management and referral criteria of adult patients with newly diagnosed hypothyroidism.

Background

Hypothyroidism affects up to 4.6% of the UK population and is especially prevalent in elderly females. Presentation of thyroid dysfunction may be non-specific and linked to other pathologies e.g. iron deficient anaemia, hypercholesterolemia. Long-term effects of hypothyroidism include reduced cardiac output, anaemia, hypogonadism, depression, mental confusion and increased susceptibility to infections.

When is hypothyroidism suspected?

Patients may be asymptomatic especially if they have subclinical hypothyroidism.

The number and severity of symptoms will be dependent on the degree of hypothyroidism.

Symptoms include;

- Weakness and tiredness
- Constipation
- Weight gain
- Dry Skin
- Bradycardia
- Irregular periods/ infertility
- Depression / unexplained anxiety

Thyroid function testing

- Unless secondary hypothyroidism is suspected, first line testing is TSH alone
- If TSH is raised the laboratory automatically measures Free T4

Do not test for thyroid dysfunction during an acute illness unless you suspect the acute illness is due to thyroid dysfunction, because the acute illness may affect the test results.

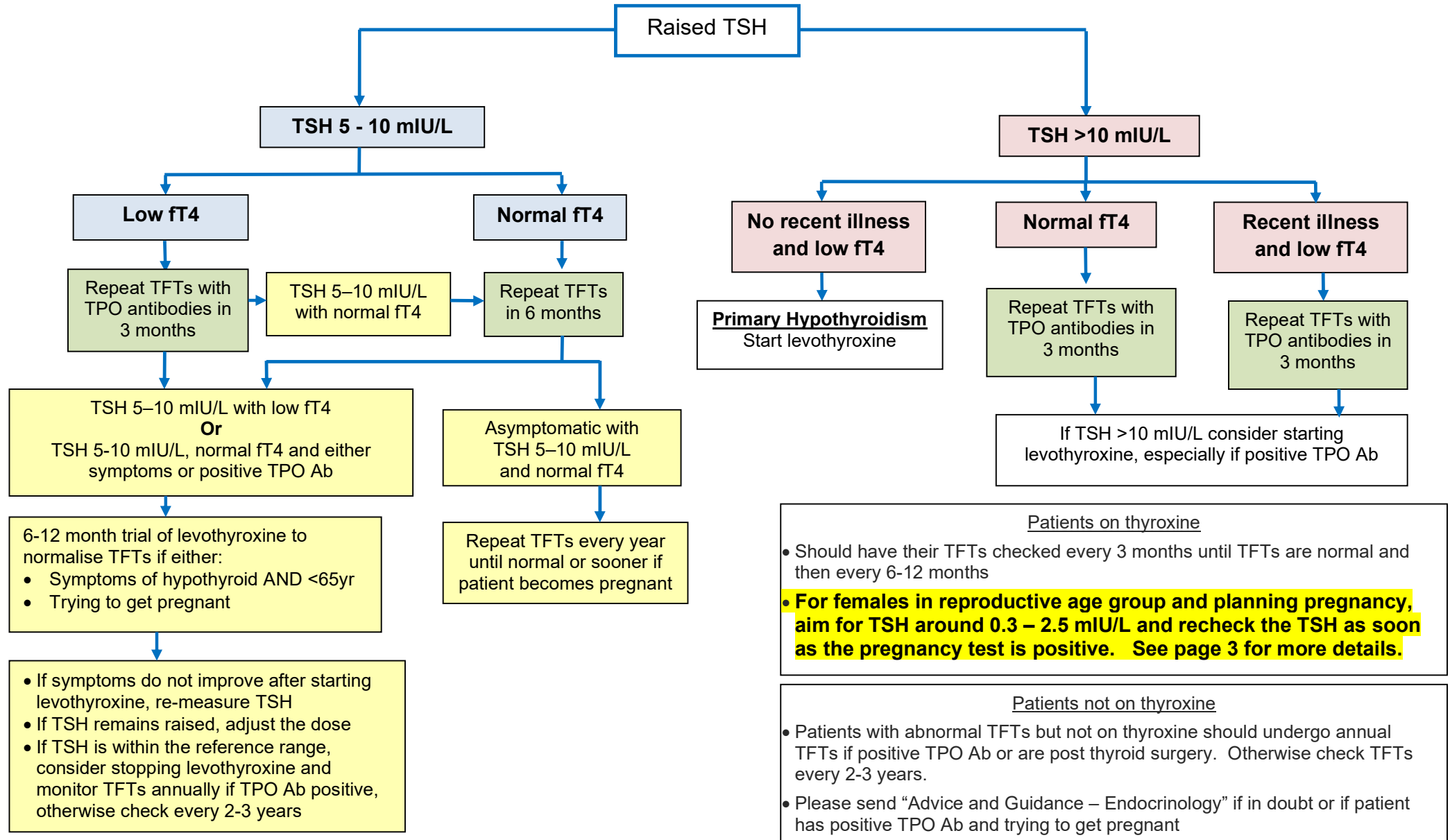
Consider repeating TFTs if symptoms worsen or new symptoms develop
(but no sooner than 6 weeks from the most recent test).

Definition

- Primary hypothyroidism - Increased TSH and low free T4
- Always consider pituitary disease (secondary hypothyroidism) if normal TSH and low FT4 and FT3. This can be explored through an Endocrine Advice & Guidance.

What are the major causes of hypothyroidism?

- Auto-immune thyroiditis e.g. Hashimoto's thyroiditis
- Iodine deficiency
- Drugs, especially Lithium and Amiodarone
- Post Thyroidectomy/Radio-iodine treatment



Hypothyroidism – what to do next:

Hypothyroidism is defined as a high TSH and low thyroid hormones.

- Diagnosis ?autoimmune, transient thyroiditis
- Check drugs e.g. Amiodarone and Lithium
- Check for family history
- Examine for goitre

Subclinical hypothyroidism

Subclinical hypothyroidism is defined as a raised TSH and normal thyroid hormones. This may be transient and due to non-thyroidal illness. These patients should have a repeat test to monitor thyroid status after acute illness. If TSH result remains borderline suggest review using the following checklist:

- Check drugs e.g. Amiodarone and lithium
- Check thyroid peroxidase antibodies (TPO) if not already carried out as a high level of TPO antibodies increases the risk of progression to overt hypothyroidism
- Consider a trial of treatment if the patient is symptomatic <65years, after excluding other causes
- Consider co-existing pathology e.g. hypertension/cardiac disease before starting treatment especially in the elderly
- Positive TPO antibodies increase the risk of progression to overt hypothyroidism by 2.1% per annum in females, higher in males
- If the status is proceeded by treatment with radioiodine, early start of thyroxine recommended

Following a trial of levothyroxine: If the patient remains symptomatic despite a normalised TSH to within the reference range with levothyroxine, conclusion can be made that thyroid pathology is not the cause of the symptoms and stop levothyroxine. Monitor TFTs annually if TPO Ab positive, otherwise check every 2-3 years.

Starting patients on Thyroxine

Please refer to the NICE guideline on Hypothyroidism but using an individualised patient approach and taking into account the thyroid hormone levels

<https://cks.nice.org.uk/topics/hypothyroidism/prescribing-information/levothyroxine/>

Thyroid monitoring of patients on Thyroxine

The aim of treatment should be to restore and maintain the TSH level within the reference range. Once started on thyroxine, TSH can take up to 4-6 months to recover. It is recommended to repeat TSH after 6-8 weeks to ensure adequate titration of thyroxine dose. Once TSH is stable it is recommended to carry out annual monitoring.

Considerations if unable to normalise TSH

Common causes of persistently elevated TSH in patients on levothyroxine replacement are:-

- Poor compliance with thyroxine (major cause)
- Inadequate thyroxine dose
- Interaction with drugs that reduce thyroxine absorption/action
- Taking food with levothyroxine
- Malabsorption
- Coeliac disease or autoimmune gastritis
- Laboratory assay interference



If compliance and drug interactions have been excluded, discuss with duty biochemist or consultant endocrinologist for further investigations/referral.

For a list of drugs that affect thyroxine adsorption/action see appendix 1.

Thyroid monitoring in pregnancy

- Routine screening for primary thyroid disease in pregnancy is not recommended particularly in the first trimester when suppression of TSH due to hCG effects causes apparent thyrotoxic results (but normal for pregnancy)
- **Enquire about pregnancy plans when the patient attends for the annual review of their thyroid function.** The foetus is dependent on maternal thyroxine in the first 12 weeks of life, and thyroxine requirements increase at the time of conception so it is important that the woman's thyroid function is optimised preconceptually. **Treated hypothyroid women planning pregnancy should have a TSH of 0.3 – 2.5 mIU/L.**
- Hypothyroid women, upon confirmation of pregnancy, should independently increase their thyroxine dose by 25-30% and contact their GP to arrange an urgent TSH test
- All pregnant hypothyroid women should be referred to the antenatal clinic. In stable hypothyroidism, TSH should be monitored approximately every 8 weeks. All those with unstable hypothyroidism as well as thyroxine-treated hypothyroid women who previously underwent thyroidectomy or radioactive iodine treatment for thyrotoxicosis or thyroid cancer will be further referred to a consultant endocrinologist.
- When requesting thyroid function tests, please state the gestation, so the lab can quote the pregnancy and gestation specific normal ranges

Myxoedema coma

In modern times this type of presentation is rare and the diagnosis is usually made before biochemical results are available. Patients should be transferred to secondary care for appropriate assessment and management of the disease.

A final point

The thyroid hormone results should always be interpreted with clinical symptoms, age and concurrent disease.

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131
Endocrinology Advice (urgent)	07879 115507 (9am to 5pm, Mon – Fri)
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
Endocrinology Advice (urgent)	01246 513104
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Burton

Duty Biochemist	01283 593313 (8am - 4 pm, Mon-Fri)
On Call Consultant Biochemist	Via QHB switchboard, 01283 566333

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Endocrinology Advice (urgent)

Endocrinology advice (non-urgent) Choose 'Advice and Guidance' from NHS e-Referrals

Patient information sheet

Patient information sheets can be obtained from the British Thyroid Foundation:

<http://www.btf-thyroid.org>

References

1. NICE guideline NG145 (Nov 2019, updated Oct 2023). Thyroid disease: assessment and management. <https://www.nice.org.uk/guidance/ng145>
2. UK Guidelines for Thyroid Function Tests (ACB, British Thyroid Association and British Thyroid Foundation, 2006). https://www.british-thyroid-association.org/sandbox/bta2016/uk_guidelines_for_the_use_of_thyroid_function_tests.pdf
3. Chakera A et al, Treatment for primary hypothyroidism: current approaches and future possibilities. Drug design, development and therapy 2012;6, 1-11.
4. Dayan CM, Interpretation of thyroid function tests. Lancet, 357, Feb 24, 2001, 619-622.

Appendix 1

Drugs known to affect the absorption of thyroxine. These should not be taken at the same time as the thyroxine dose. Discuss with consultant endocrinologist for more information.

- Iron
- Anti-acids e.g. calcium carbonate
- Aluminium hydroxide
- Zinc/Magnesium salts
- High dose iodine
- High fibre

Drugs known to increase the metabolism of thyroxine

- Phenytoin
- Carbamazepine
- Phenobarbital

Drugs known to decrease the metabolism of thyroxine

- Warfarin
- Amitriptyline

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	V2 May 2015	Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	Routine Review
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Approval	Date	Name(s)	Designation
Secondary Care UHDB	27/08/2025	Dr A Ugur, Dr R Stanworth, Dr H Ali	Consultant Endocrinologists
Secondary Care CRH	03/09/2025	Dr P Shillo	Consultant Endocrinologist
Primary Care	04/09/2025	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	12/09/2025	Ms S Knowles	Consultant Clinical Scientist